1 Some kind of Generalization

The Binomial, Poisson and the Negative Binomial Distribution can be expressed in the following form.

$$\frac{f(x+1)}{f(x)} = \frac{\alpha+\beta x}{(x+1)} \quad , x = 0, 1, 2, \ldots.$$

where, α & β are constants, $\beta \neq 1$.
First, we find the α, β for every distribution.
(1) $\frac{f(x+1)}{f(x)} = \frac{\binom{n}{x+1}p^{x+1}q^{n-x-1}}{\binom{n}{x}p^x q^{n-x}}$ If $X \sim \text{Bin}(n,p)$.

$$= \frac{n!}{(x+1)!(n-x-1)!}\frac{x!(n-x)!}{n!}p/q$$
$$= \frac{(n-1)}{(x+1)}\frac{p}{q}$$

So, $\alpha + \beta x = (n-1)\frac{p}{q} = \frac{np}{q} - \frac{p}{q}x$

$$\therefore \alpha = \frac{np}{q}, \beta = -\frac{p}{q}$$

(2) If $X \sim \text{Poi}(\lambda)$, $\frac{f(x+1)}{f(x)} = \frac{e^{-\lambda}\lambda^x/x!}{e^{-\lambda}\lambda^x/x!} = \frac{\lambda}{(x+1)}$
So, $(\alpha + \beta x) = \lambda$
$\therefore \alpha = \lambda, \beta = 0$
(3) If $X \sim \text{NB}(r,p)$,

$$\frac{f(x+1)}{f(x)} = \frac{\binom{x+r}{x+1}p^r q^{x+1}}{\binom{x+r-1}{x}p^r q^x}$$
$$= \frac{\frac{(x+r)!}{(x+1)!(r-1)!}}{\frac{(x+r-1)!}{(x)!(r-1)!}}q = \frac{x+r}{x+1}q$$

So, $(\alpha + \beta x) = (x+r)q$
$\Rightarrow q = rq, \beta = q$
Using the above type finding mean and variance,

$$\frac{f(x+1)}{f(x)} = \frac{\alpha+\beta x}{(x+1)}$$
$$\Rightarrow (x+1)f(x+1) = (\alpha+\beta x)f(x) = \alpha f(x) + \beta x f(x)$$
$$\Rightarrow \sum_{x=0}^{\infty}(x+1)f(x+1) = \sum_{x=0}^{\infty}\alpha f(x) + \sum_{x=0}^{\infty}\beta x f(x)$$
$$\Rightarrow \sum_{y=x+1=1}^{\infty} y f(y) = \alpha \cdot 1 + \beta E(X) \Rightarrow E(Y) = \alpha + \beta E(X)$$
$$\Rightarrow E(X) = \alpha + \beta(x) \Rightarrow E(X) = \frac{\alpha}{1-\beta}$$

(1) For $\text{Bin}(n,p), E(X) = \frac{np/q}{1+\frac{p}{q}} = \frac{np}{1} = np$
(2) For, $\text{Poi}(\lambda)$, $E(X) = \frac{\lambda}{1+0} = \lambda$

(3) For, NB$(r,p), E(X) = \frac{rq}{1-q} = \frac{rq}{p}$
Again,

$$(x+1)f(x+1) = \alpha f(x) + \beta x f(x)$$

$$\Rightarrow \sum_{x=0}^{\infty} x(x+1)f(x+1) = \sum_{x=0}^{\infty} \alpha x f(x) + \sum_{x=0}^{\infty} \beta x^2 f(x)$$

$$\Rightarrow \sum_{x+1-y=1}^{\infty} (y-1)yf(y) = \alpha E(X) + \beta E\left(X^2\right)$$

$$\Rightarrow \sum_{y=1}^{\infty} y^2 f(y) - \sum_{y=1}^{\infty} yf(y) = \alpha E(X) + \beta E\left(X^2\right)$$

$$\Rightarrow \sum_{x=0}^{\infty} x^2 f(y) - \sum_{x=0}^{\infty} xf(x) = \alpha E(X) + \beta E\left(X^2\right)$$

$$\Rightarrow E\left(X^2\right) - E(X) = \alpha E(X) + \beta E\left(X^2\right)$$

$$\Rightarrow E\left(X^2\right) = \frac{E(X)(\alpha+1)}{(1-\beta)} = \frac{\alpha(\alpha+1)}{(1-\beta)^2} \quad [\text{ Pulting value of } E(x)].$$

$$\therefore \text{Var}(X) = \frac{E\left(X^2\right) - [E(X)]^2}{(1-\beta)^2} - \frac{\alpha^2}{(1-\beta)^2} = \frac{\alpha^2+\alpha-\alpha^2}{(1-\beta)^2} = \frac{\alpha}{(1-\beta)^2}$$

(1) For Bin(n,p), var$(x) = \frac{np/q}{\left(1+\frac{p}{q}\right)^2} = \frac{\frac{np}{q}}{\frac{(q+p)^2}{q^2}} = npq$
(2) For Poi(λ), var$(x) = \frac{\lambda}{(1-0)^2} = \lambda$
(3) For NB(r, p) var$(x) = \frac{rq}{(1-q)^2} = \frac{rq}{p^2}$

Mean Deviation:

$$E[X - \mu l\ |] = 2\sum_{x=0}^{[\mu]} (\mu - x)f(x)$$

$$= 2\sum_{x=0}^{[\mu]} \left\{ \left(\frac{\alpha}{1-\beta} \right) - x \right\} f(x)$$

$$= 2\sum_{x=0}^{[\mu]} \left(\frac{\alpha - x + \beta x}{1-\beta} \right) f(x)$$

$$= \frac{2}{(1-\beta)} \sum_{x=0}^{[\mu]} \{(\alpha + \beta x)f(x) - xf(x)\}$$

$$= \frac{2}{1-\beta} \sum_{x=0}^{[\mu]} [(x+1)f(x+1) - xf(x)]$$

$$= \frac{2}{1-\beta} [([u]+1)f\{[u]+1\}]$$

(1) For Bin (n,p), MD$_\mu = \frac{2}{1+\frac{p}{q}} [([\mu]+1)f\{[\mu]+1\}]$

$$= 2q[([\mu]+1)f\{[\mu]+1\}]$$

(2) For Poi (λ), MD$_\mu = \frac{2}{1+0} [([\mu]+1)f\{[\mu]+1\}]$

$$= 2[([\mu]+1)f\{[\mu]+1\}]$$

(3) For NB(n,p) MD$_\mu = \frac{2}{1-q} [([\mu]+1)f\{[\mu]+1\}]$

$$= \frac{2}{p} [([\mu]+1)f\{[\mu]+1\}]$$

- Similarly, if X has a discrete distribution with mass points $0, 1, 2, \ldots$ and with pmf $f(x)$, such that $\frac{f(x)}{f(x-1)} = \frac{\alpha+\beta x}{x}$ for $x = 1, 2, 3, \ldots$. Then, find $E(X), V(X)$, MD_μ etc.

  First we find α and β for every distribution.

  (1) For $X \sim \text{Bin}(n, p)$

$$\begin{aligned}\frac{f(x)}{f(x-1)} &= \frac{\binom{n}{x}p^x q^{n-x}}{\binom{n}{x-1}p^{x-1}q^{n-x+1}} \\ &= \frac{\frac{n!}{x!(n-x)!}p}{\frac{n!}{(n-x)!(n-x-1)!}q} = \frac{(n-x+1)}{x}\frac{p}{q}\end{aligned}$$

$$\therefore \quad \alpha + \beta x = (n-x+1)\frac{p}{q} = (n+1)\frac{p}{q} - x\frac{p}{q}$$

$$\therefore \quad \alpha = (n+1)\frac{p}{q}, \beta = -\frac{p}{q}.$$

(2) For $X \sim \text{Poi}(\lambda)$, $\frac{f(x)}{f(x-1)} = \frac{e^{-\lambda}\lambda^x/x!}{e^{-\lambda}\lambda^{x-1}/(x-1)!} = \frac{\lambda^x}{\lambda!}\frac{(x-1)!}{\lambda^{x-1}} = \frac{\lambda}{x}$

$$\begin{aligned}&\therefore \quad \alpha + \beta x = \lambda \\ &\Rightarrow \quad \alpha = \lambda, \beta = 0.\end{aligned}$$

(3) For $X \sim \text{NB}(n, p)$,

$$\begin{aligned}\frac{f(x)}{f(x-1)} &= \frac{\binom{x+r-1}{x}p^r q^x}{\binom{x+r-2}{x-1}p^r q^{x-1}} \\ &= \frac{\frac{(x+r-1)!}{x!(r-1)!}}{\frac{(x+r-2)!}{(x-1)!(r-1)!}}q \\ &= \frac{x+r-1}{x}q\end{aligned}$$

$\therefore \alpha + \beta x = (x+r-1)q \quad \Rightarrow \alpha = (r-1)q, \beta = q$

2 Negative Binomial and Poisson Distribution

a) In practice, a great care has to be taken to avoid erroneous application, of Poisson distribution to counting variables.

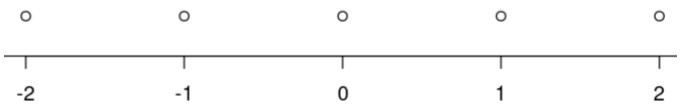
Example: In studying the distributions of deaths on insects, the Poisson Distribution is not applicable/appropriate, as the deaths of insects form cluster which are inconsistent with the assumptions of Poisson process. Under the assumptions of Poisson process, events are uniformly distributed over time, space. The Negative Binomial distribution can be obtained as are a result of clustering of events over time, space.

i) The number of insects bites in an apple.

ii) The number of accidents on an individual working in an explosive shell.

iii) The number of deaths in an epidemic in a region.

b) (i)



$$\bar{x} = 0 \cdot \sum_{i=1}^{5} x^2 = 2\left\{1^2 + 2^2\right\}$$

(ii)



In (i), values are uniformly distributed but in (ii) values form clusters.

Note that, variance (data I) $<$ variance (data II)

Values form cluster $\Rightarrow$ variance is large.

$\Rightarrow$ NB is suitable. Given a data on a counting RV if $\bar{x} \leq S^2$, then Poisson laws are appropriate but $\bar{x} \ll s^2$, then NB law is appropriate.